Junghoon Chung

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Summary

Ph.D. student in Industrial & Systems Engineering specializing in machine learning, causal inference, and multimodal human-performance modeling. Builds reproducible data pipelines and evaluation frameworks that translate physiological, behavioral, and contextual data into interpretable, adaptive systems. Seeking Data Science or Applied Scientist internship opportunities.

Education

University of Washington, Seattle Sep 2025 – Jun 2029 (Expected)
Ph.D. in Industrial and Systems Engineering; Dean's Fellowship Recipient

Virginia Tech, Blacksburg Aug 2023 – May 2025
M.S. in Industrial and Systems Engineering (Thesis Track), GPA: 3.89/4.0

Yonsei University, Seoul Mar 2017 – Feb 2023
B.S. in Industrial Engineering, GPA: 3.52/4.0; Student Council President

Experience

Graduate Research Assistant, HAS Lab, University of Washington Sep 2025 – Present

- Build machine-learning models of vigilance using multimodal physiological time-series data and preprocessing pipelines for heterogeneous sensor streams in Python and scikit-learn.
- Engineer 30+ physiological and contextual features across 10,000+ observations per participant; feature refinement improved predictive performance by approximately 20%.

Graduate Research Assistant, OEB Lab, Virginia Tech Aug 2023 – May 2025

- Developed end-to-end pipelines to collect, preprocess, and analyze multimodal datasets spanning EMG, EEG, IMU, motion capture, behavioral logs, and survey data.
- Applied zero-shot natural-language classification, time-series forecasting, data segmentation, and clustering to longitudinal workplace data with 100,000+ records per participant.

Selected Projects

Human Vigilance Modeling with Causal Inference *University of Washington*
Built and compared alertness models using multimodal physiological data, including ECG, blood-volume pulse, electrodermal activity, heart rate, and eye tracking. Evaluated time-series aggregation windows from 5 to 60 seconds and examined the relationship between physiological signals and task reaction time.

Real-Time Vigilance Monitoring Interface *University of Washington*
Developing an EmotiBit-based pipeline that streams heart rate, electrodermal activity, and photoplethysmography into a trained model for continuously updated vigilance estimates. Exploring live-learning updates to personalize predictions as individual data accumulate.

Driver Response Modeling for Vehicle-to-Everything Pedestrian Alerts *City of Bellevue / University of Washington*
Designing a field study that integrates vehicle kinematics, electrodermal activity, heart rate, blood-volume pulse, workload ratings, and interviews to model driver responses to smartphone pedestrian alerts.

Phase-Based Modeling of Automated-Vehicle Takeover *University of Washington*
Developing and evaluating phase-specific behavioral features for pre-, during-, and post-takeover periods to test whether temporal modeling improves prediction beyond aggregate measures.

Context-Aware Sit-Stand Intervention *Virginia Tech*
Segmented three weeks of longitudinal behavioral data with more than 100,000 records per participant into personalized usage clusters, then fit time-series models to forecast posture transitions.

Skills

Programming: Python, SQL (MySQL), R, Swift (watchOS)
Machine Learning: scikit-learn, PyTorch, TensorFlow, causal inference, feature engineering
Data Analysis: Statistical modeling, time-series analysis, multimodal sensing, experimental design
Communication: Technical reporting, cross-functional collaboration

Awards

Dean's Fellowship, College of Engineering — University of Washington, 2025–2029
Social-Innovation Capability Video Contest, 2nd Place — Yonsei University, 2022
Merit-based Scholarship (Academic Excellence) — Yonsei University, 2021